

Problem 41 (Dutch IMO TST 2016 Day 2 Problem 1). Let a, b, c be positive real numbers. Prove that

$$a + \sqrt{ab} + \sqrt[3]{abc} \leq \frac{4}{3}(a + b + c).$$

Solution. By the AM-GM inequality, we have

$$\begin{aligned} \frac{a}{4} + b &\geq 2\sqrt{\frac{a}{4} \cdot b} = \sqrt{ab}, \\ \frac{a}{12} + \frac{b}{3} + \frac{4c}{3} &\geq \sqrt[3]{\frac{a}{4} \cdot b \cdot 4c} = \sqrt[3]{abc}. \end{aligned}$$

Adding these and adding a on both sides, we obtain the desired inequality. Equality holds when $a : b : c = 16 : 4 : 1$. $\square$

Problem 42 (Irish MO 2020). Let a, b, c be positive real numbers. Prove that

$$\sqrt[7]{\frac{a}{b+c} + \frac{b}{c+a}} + \sqrt[7]{\frac{b}{c+a} + \frac{c}{a+b}} + \sqrt[7]{\frac{c}{a+b} + \frac{a}{b+c}} \geq 3.$$

Solution. By the AM-GM inequality, we have

$$\sqrt[7]{\frac{a}{b+c} + \frac{b}{c+a}} + \sqrt[7]{\frac{b}{c+a} + \frac{c}{a+b}} + \sqrt[7]{\frac{c}{a+b} + \frac{a}{b+c}} \geq 3 \sqrt[21]{\left(\frac{a}{b+c} + \frac{b}{c+a}\right) \left(\frac{b}{c+a} + \frac{c}{a+b}\right) \left(\frac{c}{a+b} + \frac{a}{b+c}\right)}.$$

So it suffices to prove $\left(\frac{a}{b+c} + \frac{b}{c+a}\right) \left(\frac{b}{c+a} + \frac{c}{a+b}\right) \left(\frac{c}{a+b} + \frac{a}{b+c}\right) \geq 1$.

WLOG assume $a \geq b \geq c$. Then $\frac{1}{b+c} \geq \frac{1}{c+a} \geq \frac{1}{a+b}$. By Chebyshev's inequality and the AM-GM inequality, we have

$$\frac{a}{b+c} + \frac{b}{c+a} \geq \frac{1}{2}(a+b) \left(\frac{1}{b+c} + \frac{1}{c+a}\right) \geq (a+b) \sqrt{\frac{1}{(b+c)(c+a)}}.$$

Similarly, we have

$$\begin{aligned} \frac{b}{c+a} + \frac{c}{a+b} &\geq (b+c) \sqrt{\frac{1}{(c+a)(a+b)}}, \\ \frac{c}{a+b} + \frac{a}{b+c} &\geq (c+a) \sqrt{\frac{1}{(a+b)(b+c)}}. \end{aligned}$$

Multiplying these three inequalities, we obtain our desired inequality. So we are done. Equality holds when $a = b = c$. $\square$

Problem 43 (Dutch IMO TST 2017 Day 3 Problem 2). Let $a_1, a_2, \dots, a_n$ be real numbers such that $a_1 + a_2 + \dots + a_n = 0$. Define

$$b_k = a_1 + a_2 + \dots + a_k$$

for $k = 1, 2, \dots, n$. It is known that $b_i(a_{j+1} - a_{i+1}) \geq 0$ for all $1 \leq i \leq j \leq n-1$. Prove that

$$\max\{|a_1|, |a_2|, \dots, |a_n|\} \geq \max\{|b_1|, |b_2|, \dots, |b_n|\}$$

Solution. Let $|b_k| = \max\{|b_1|, |b_2|, \dots, |b_n|\}$. By flipping the signs of all terms if necessary, we may assume $b_k \geq 0$.

If $a_k \geq b_k$, then $\max\{|a_1|, |a_2|, \dots, |a_n|\} \geq a_k \geq b_k$ and we are done.

If $a_k < b_k$, then $k \geq 2$ (since $b_1 = a_1$) and $b_{k-1} = b_k - a_k > 0$. Also, we have $a_k = b_k - b_{k-1} \geq 0$ since otherwise $|b_{k-1}| > |b_k|$.

Using the given condition with $i = k-1$, we have $a_{j+1} - a_k \geq 0$ for any $j \geq k-1$. This means $a_j \geq a_k \geq 0$ for any $j \geq k$. But then we obtain the contradiction

$$0 = a_1 + a_2 + \dots + a_n = b_{k-1} + a_k + a_{k+1} + \dots + a_n > 0.$$

Therefore, the assertion is proved. $\square$

Problem 44 (Polish MO 2018 Final Problem 3). Find all real numbers c for which there exists a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + f(y)) + cxy = f(x + y) \quad (1)$$

for any $x, y \in \mathbb{R}$.

Solution. When $c = 0$, we can take the zero function as a solution since both sides become 0 in that case. We now show that there is no solution when $c \neq 0$.

Putting $y = -x$ in (1), we obtain $f(f(x) + f(-x)) = f(0) + cx^2$. The right-hand side can take any value larger than $f(0)$ if $c > 0$ (or any value smaller than $f(0)$ if $c < 0$). Therefore, $(f(0), \infty)$ (or $(-\infty, f(0))$) is a subset of the range of f .

Putting $y = 0$ in (1), we obtain $f(f(x) + f(0)) = f(x)$. This implies $f(t + f(0)) = t$ for any t in the range of f . In particular, this means $f(s) = s - f(0)$ for any sufficiently large (or sufficiently small) s .

Considering x and y in (1) which are sufficiently large (or sufficiently small), the terms $x + y - 2f(0)$ and $x + y$ are also sufficiently large (or sufficiently small). Therefore, (1) becomes $f(x - f(0) + y - f(0)) + cxy = x + y - f(0)$, i.e.

$$x + y - 3f(0) + cxy = x + y - f(0).$$

This implies $cxy = 2f(0)$, which cannot hold for sufficiently large (or sufficiently small) x and y . Therefore, there is no such function f if $c \neq 0$. □